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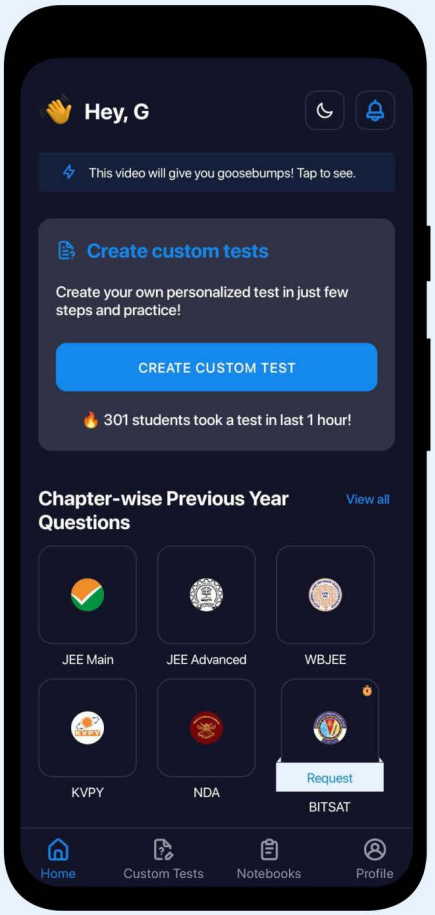

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STOICHIOMETRY

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2 STOICHIOMETRY

2.1 STOICHIOMETRY :

Derived from the Greek word "stoichio" and "metry", Stoichiometry literally translates to Measurement of elements. It deals with determination of quantities of products or reactants of a chemical reaction.

Stochiometry can be divided into two sub-categories :

(a) Gravimetric analysis

(b) Volumetric analysis.

There are two methods to solve problems in stochiometry, one using the mole concept while the other using equivalent concept. Both methods are equally important and would be discussed in the text below.

Let a balanced equation

$$mA + nB \rightarrow pC + qD$$

In this reaction

A and B are the reactants

C and D are the products

m, n p and q are stoichiometric coefficient.

Mole concept states that

m moles of A = n moles of B = p moles of C = q moles of D.

2.2 CHEMICAL EQUATIONS AND STOICHIOMETRY

Let the balanced chemical equation we have is

$$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$$

The quantitative information drawn from this balanced chemical equation is

(a) The molar ratio in which two reactants (MnO_2 and HCl) reacting is 1 : 4.

(b) The molar ratio between two products can be known i.e. moles of H_2O produced would be double the moles of $MnCl_2$ produced.

(c) The initial moles of MnO_2 and HCl (to be taken in vessel) for the reaction to occur may not necessarily be 1 and 4 respectively or also should not be in the molar ratio of 1 : 4.

(d) We can start reaction with MnO_2 and HCl taken in any molar ratio, but the moles of two reacting will always be in the molar ratio of 1 : 4.

(e) The balanced chemical equation should follow the law of conservation of mass.

Let us consider the same chemical system with initial composition (in terms of mole) as $n^0_{MnO_2}$, $n^0_{MnCl_2}$, $n^0_{H_2O}$ and $n^0_{Cl_2}$. The n^0_{HCl} is four times of $n^0_{MnO_2}$. When the reaction occurs, these mole numbers change as the reaction progresses. The mole numbers of the various species do not change independently but the change are related by the stoichiometric coefficients in the chemical equation. Let after time 't' from the commencement of reaction, the moles of MnO_2 reacting be x, then the moles of HCl reacting in the same time interval be 4x since MnO_2 and HCl react in the molar ratio of 1 : 4.

Thus, after time t, the composition of the system would be

$$n_{MnO_2} = n^0_{MnO_2} - x$$

$$n_{HCl} = n^0_{HCl} - 4x$$

$$n_{MnCl_2} = n^0_{MnCl_2} + x$$

$$n_{H_2O} = n^0_{MnO_2} + 2x$$

$$n_{Cl_2} = n^0_{Cl_2} + x$$

The algebric signs, + and – indicates that the reactants are consumed and products are produced.

After long time interval from the commencement of reaction i.e. after ∞ time, the composition of the system would be

$$n_{MnO_2} = 0,\ n_{HCl} = 0$$

$$n_{MnCl_2} = n^0_{MnCl_2} + n^0_{MnO_2} = n^0_{MnCl_2} + \frac{n^0_{HCl}}{4}$$

$$n_{H_2O} = n^0_{H_2O} + 2n^0_{MnO_2} = n^0_{H_2O} + \frac{n^0_{HCl}}{2}$$

$$n_{Cl_2} = n^0_{Cl_2} + n^0_{MnO_2} = n^0_{Cl_2} + \frac{n^0_{HCl}}{4}$$

Example 1

Calculate the weight of time (CaO) that can be prepared by heating 200 kg of limestone ($CaCO_3$) which is 95% pure.

Solution :

$$CaCO_3 \rightarrow CaO + CO_2$$

$$\text{Actual amont of } CaCO_3 = \frac{95}{100} \times 200000\text{g} = 190000\text{g}.$$

Since Ca atoms are conserved,

∴ moles of Ca in $CaCO_3$= moles of Ca in CaO

$$1 \quad \text{moles of } CaCO_3 = 1 \quad \text{moles of CaO}$$

$$\frac{190000}{100} = \frac{\text{Wt. of CaO}}{56}$$

$$\text{Wt. of CaO} = 106400 = 106.4 \text{ kg}$$

2.3 CONCEPT OF LIMITING REAGENT

In the single-reactant reactions, the calculations are carried out with only that amount of the reactant which has converted to the product.

In the reactions where more than one reactant are involved, one has first to identify the **limiting reactant, i.e., the reactant which is completely consumed.** All calculations are to be carried out with the amount of the limiting reactant ony. **Now the question is how to know the limiting reactant?** The procedure is simple as illustrated below :

Initially ...	5 mole		12 mole		0 mole
	A	+	2 B	→	4 C

If A is the limiting reactant : Moles of C produced = 20

If B is the limiting reactant : Moles of C produced = 24

The reactant producing the least number of moles of the product is the limiting reactant and hence A is the limiting reactant. Thus;

Initially ...	5 mole		12 mole		0 mole
	A	+	2 B	→	4 C
Finally ...	0 mole		2 mole		20 mole

The limiting reactant can also be ascertained by the knowing the initial number of equivalents (or milli-equivalents) of each reactant. **The reactant with the least number of equivalents (or milli-equivalents) is the limiting reactant.** The equivalent method to identify the limiting reactant can be applied even if the chemical equation is not balanced.

Suppose you are given 5 moles of MnO_2 and 18 moles of HCl and you have to calculate the maximum amount of $MnCl_2$ being produced.

The balanced equation is :

$$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$$

initial moles	5	18

Now since 1 mole of MnO_2 reacts completely with 4 mole of HCl to produce 1 mole of $MnCl_2$. Therefore, 5 mole of $MnCl_2$ will react with 20 mole of HCl. But since moles of HCl is only 18. So, HCl will be the reagent, which will be consumed first, and hence HCl is the limiting reagent. If MnO_2 have been the limiting reagent 5 mole of it would have given 5 moles of $MnCl_2$ but it is not so. As one can see from the balanced chemical equation that 1 mole of $MnCl_2$ is produced from 4 mole

of HCl. So 18 moles of HCl will produce 4.5 moles of $MnCl_2$. So, one can define limiting reagent.

"The reagent producing the least number of moles of products is the limiting reagent."

YIELD IN A REACTION WITH LIMITING REAGENT

For reactions where one of the reagent is limiting, the product yield would not be 100% but would depend on amount of limiting reagent.

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{Theoretical maximum yield}} \times 100$$

$$\therefore \quad \%\ \text{yield of the product} = \frac{\text{actual yield}}{\text{Theoretical maximum yield}} \times 100$$

Example 2

The reaction, $2C + O_2 \rightarrow 2CO$ is carried out by taking 24 g of carbon and 96 g O_2, find out :

(a) Which reactant is left in excess ?

(b) How much of it is left ?

(c) How many mole of CO are formed ?

(d) How many gm of other reactant should be taken so that nothing is left at the end of reaction ?

Solution :

	$2C$	+	O_2	$\rightarrow$	$2CO$
Mole before reaction	$\frac{24}{12}$		$\frac{96}{32}$		
	$= 2$		$= 3$		0
Mole after reaction	0		2		2

(a) O_2 is left in excess.

(b) 2 mole of O_2 or 64 gm O_2 is left

(c) 2 moe of CO or 56 gm CO is formed

(d) To use O_2 completely total 6 mole of carbon or 72 gm carbon is needed.

2.4 GRAVIMETRIC ANALYSIS

Gravimetric analysis is an analytical technique based on the measurement of mass of solid substances and/or volume of gaseous species.

The gravimetric analysis is broadly classified into three heads.

(a) Mass-mass relationship

(b) Mass-volume relationship and

(c) Volume-volume relationship

2.4.1 Mass–Mass Relationship

This relates the mass of a species (reactant or product) with the mass of another species (reactant or product) involved in a chemical reaction.

Let us consider a chemical reaction

$$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$$

Let the mass of $CaCO_3$ taken be x g and we want to calculate the mass of CaO obtained by heating x g $CaCO_3$. Then the moles of $CaCO_3$ taken would be $\frac{x}{M_1}$ (where M_1 represents the molar mass of $CaCO_3$). According to the balanced reaction, the molar ratio of $CaCO_3$ and CaO is 1 : 1. So same number of moles $\left(\frac{x}{M_1}\right)$ of CaO would be formed. Now for converting the moles of CaO into mass of CaO obtained, we need to multiply the moles of CaO with the molar mass of CaO. Let the molar mass of CaO be M_2, so the mass of CaO obtained by heating x g of $CaCO_3$ would be $\left(\frac{x}{M_1} \times M_2\right)g$.

Example 3

What will be the mass of CaO produced by 10 gm of $CaCO_3$. The reaction is

$$\mathbf{CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)}$$

Solution :

	$CaCO_3(s)$	$\xrightarrow{\Delta}$ $CaO(s)$	$+ CO_2(g)$
at t = 0	10 gm	0	0
at t = ∞	0 (completely consumed)	?	?

$$\text{No. of moles of } CaCO_3 = \frac{\text{wt. in gm}}{\text{Mol. wt}} = \frac{10}{100}$$

$$= 0.1 \text{ mole}$$

from mole concept

$$1 \text{ mole of } CaCO_3 = 1 \text{ mole of CaO}$$

⇒ moles of CaO produced will be equal to moles of $CaCO_3$

$$\text{moles of CaO(s)} = 0.1 \text{ mole}$$

$$\text{wt. of CaO} = 0.1 \quad \text{mol. wt. of CaO}$$

$$= 0.1 \quad 56$$

$$= 5.6 \text{ gm}$$

2.4.2 Mass-Volume Relationship

This establishes the relationship between the mass of a species (reactant or product) and the volume of a gaseous species (reactant or product) involved in a chemical reaction. Let us take x g of $CaCO_3$ in a vessel of capacity V L and the vessel is heated so that $CaCO_3$ decomposes as

$$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$$

We want to find out the volume of CO_2 evolved at STP by heating x g of $CaCO_3$. Then

Moles of $CaCO_3 = \frac{x}{M_1}$ (since molar ratio of $CaCO_3$ and CO_2 is 1 : 1)

$\therefore$ Volume of CO_2 evolved at STP $= \left(\frac{x}{M_1} \times 22.4\right)L$

But, if the volume of CO_2 evolved is to be calculated at pressure P and temperature T K. Then, moles of CO_2 evolved $= \frac{x}{M_1}$

Volume of CO_2 evolved at pressure P and temperature T $= \frac{x}{M_1} \times \frac{RT}{P}$ (Using PV = nRT)

Example 4

What volume of oxygen (NTP) will be required to react with 1 gm of Ca ?

Solution :

$$Ca(s) + \frac{1}{2}O_2(g) \longrightarrow CaO(s)$$

t = 0 1 gm ?

Applying mole concept

$$1 \text{ mole of Ca} = \frac{1}{2} \text{ mole of } O_2$$

Since volume of 1 mole of O_2 (g) at NTP = 22.4 lt.

$\therefore$ volume of $\frac{1}{2}$ mole of O_2 (g) at NTP = 11.2 lt.

There can be written as

1 mole of Ca or 40 gm of Ca react with 11.2 lit. of O_2 at NTP

$\therefore$ 1 gm Ca react with $= \frac{11.2}{40} = 0.28$ lt.

2.4.3 Volume-Volume Relationship

This relationship deals with the volume of a gaseous species (reactant or product) with the volume of another gaseous species (reactant or product) involved in a chemical reaction.

Let us consider the reaction, $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$. We are given x L of N_2 at pressure P atm and temperature T K and we want to know the volume of H_2 required to react with it at the same pressure and temperature.

$$\text{Moles of } N_2 = \frac{P \times x}{RT}$$

$$\text{Moles of } H_2 \text{ required} = \frac{3 \times P \times x}{RT} \quad \text{(since molar ratio of } N_2 \text{ \& } H_2 \text{ is 1 : 3)}$$

$$\therefore \quad \text{Volume of } H_2 \text{ required at same pressure \& temperature} = \frac{3 \times P \times x \times RT}{RT \times P} = 3xL$$

This result could also have been obtained by knowing that for a gaseous relation, at the same pressure and temperature, the moles of gas is directly proportional to volume of the gas ($V \alpha n$ since P & T are constant) or molar ratio and volume ratio are same.

Thus, when x L of N_2 is taken at Pressure P atm temperature T K,then at the same pressure and temperature, the volume of H_2 required would be 3 x L (since the volume ratio of N_2 & H_2 would be same as molar ratio i.e. 1 : 3). But if the volume of H_2 required is to be calculated at another pressure P atm and temperature T'K, then

$$\text{Moles of } H_2 \text{ required} = \frac{3 \times P \times x}{RT}$$

$\therefore$ Volume of H_2 required at pressure P' atm & temperature T'K

$$= \frac{3 \times P \times x \times RT'}{RT \times P'} = \left(\frac{3x \times PT'}{P'T}\right)L$$

Example 5

8 gm of methane is burnt with 4.48 L of O_2 at STP. Find out the volume of CO_2 gas produced at STP and also the weight of CO_2 gas.

Solution :

The balanced chemical equation is

	CH_4	+	$2O_2$	$\rightarrow$	CO_2	+	$2H_2O$
	1 mol		2 mol		1 mol		
Ideal	16 gm		2 22.4 L		22.4 L		
					44 gm		

$$\text{No. of moles of } CH_4 = \frac{8}{16} = 0.5\,\text{mol}$$

$$\text{No. of moles of } O_2 = \frac{4.48\,L}{22.4\,L} = 0.2 \text{ mol}$$

Now since 1 mole of CH_4 required 2 mole (i.e., 44.82) of O_2 for complete combustion. But the given moles of O_2 is only 0.2 mol. So, O_2 is the limiting reagent.

Again, since 2 moles of O_2 reacts with 1 mole of CH_4 to give 22.4 L of CO_2 at STP.

So 0.2 mole of O_2 will react with 0.1 mol of CH_4 to give 2.24 L of CO_2.

$$\text{Wt. of } CO_2 \text{ produced} = 0.1 \text{ mol} \quad 44$$

$$= 4.4 \text{ gm of } CO_2$$

Example 6

NaCl of 95% purity is used to prepare salt cake (Na_2SO_4) by the reaction,

$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

If the product (Na_2SO_4) is only 85% pure, what weight of NaCl is used up in producing 1 kg of the impure salt cake?

Solution :

Let the weight of NaCl used up in producing 1 kg of impure product be x g.

$$\text{Mass of pure NaCl} = \frac{x \times 95}{100}$$

$$\text{Moles of pure NaCl} = \frac{x \times 95}{100 \times 58.5}$$

$$\text{Moles of pure } Na_2SO_4 = \frac{x \times 95}{100 \times 58.5} \times \frac{1}{2}$$

$$\text{Mass of pure } Na_2SO_4 = \frac{x \times 95}{100 \times 58.5} \times \frac{1}{2} \times 142 = \frac{100 \times 85}{100}$$

$\therefore$ x **= 73.72 g**

Example 7

Calculate the mass of 90% pure MnO_2 to produce 35.5 gm of Cl_2 according to following reaction.

$$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$$

Solution :

$$\underset{87\text{ gm}}{MnO_2} + 4HCl \rightarrow MnCl_2 + \underset{71\text{ gm}}{Cl_2} + 2H_2O$$

$\therefore$ 71 gm Cl_2 is produced by 87 gm of MnO_2

$\therefore$ 35.5 g Cl_2 is produced by = $\frac{87 \times 35.5}{71} = 43.5$ g MnO_2

$\therefore$ 90 gm pure MnO_2 is present in 100 gm sample

$\therefore$ 43.5 gm pure MnO_2 = $\frac{100 \times 43.5}{90} = 48.3$ gm

Example 8

$KClO_3$ decomposes on heating in two possible ways as

$$KClO_3 \xrightarrow{\Delta} KCl + 3/2\ O_2$$

$$4KClO_3 \xrightarrow{\Delta} KCl + 3KClO_4$$

when 15 g of $KClO_3$ were heated in an experiment, analysis of the product showed that 5.21 g of KCl and 4.59 g of $KClO_4$ were formed. What was the weight of $KClO_3$ remaining undecomposed ? What fraction of the $KClO_3$ decomposed yielded $KClO_4$?

Solution :

In order to calculate the weight of $KClO_3$ remaining undecomposed, we need to find out the total weight of $KClO_3$ that decomposes. The weight of $KClO_3$ that decomposes in second reaction would be known using the $KClO_4$ data. Using $KClO_4$ data, we would also calculate the weight of KCl obtained in second reaction. Since, total weight of KCl obtained is known to us, so the weight of KCl obtained in first reaction would also be known, using this we will find out the weight of $KClO_3$ decomposed in first reaction.

Moles of $KClO_4$ obtained = $\frac{4.59}{138.5}$

Moles of KCl obtained in IInd reaction = $\frac{4.59}{138.5} \times \frac{1}{3} = 0.011$ (since molar ratio of $KClO_4$ and KCl in IInd reaction is 3 : 1)

Moles of $KClO_3$ that decomposes in IInd reaction = $\frac{4.59}{138.5} \times \frac{4}{3}$ (since molar ratio of $KClO_4$ and $KClO_3$ in IInd reaction is 3 : 4)

$\therefore$ Mass of $KClO_3$ that decomposes in IInd reaction = $\frac{4.59}{138.5} \times \frac{4}{3}$ 122.5 = 5.41 g

Moles of KCl obtained in Ist reaction = $\left(\frac{5.21}{74.5} - 0.011\right) = 0.07 - 0.011 = 0.059$

Moles of $KClO_3$ that decomposes in Ist reaction = 0.059

(since molar ratio of KCl and $KClO_3$ in Ist reaction is 1 : 1)

Mass of $KClO_3$ that decomposes in Ist reaction = 0.059 122.5 = 7.23 g

---

Example 9

In a process for producing acetic acid, oxygen gas is bubbled into acetaldehyde, CH_3CHO, under pressure at 60 C in presence of suitable catalyst.

$$2CH_3CHO\ (l) + O_2\ (g) \rightarrow 2CH_3COOH\ (l)$$

In a laboratory test of this reaction, 20.0 g CH_3CHO and 10 g O_2 were put into a reaction vessel.

(a) How many grams of acetic acid can be produced by these amounts of reactants ?

(b) How many gram of the excess reactant remain after the reaction is complete ?

(c) If actual yield is 23.8 g, calculate percentage yield.

Solution :

(a) 20 g CH_3CHO = $\frac{20\,g}{44\,g\,mol^{-1}} = 0.455$ mol

10 g O_2 = $\frac{10\,g}{32\,g\,mol^{-1}} = 0.313$ mol

CH_3COOH obtained from 0.455 mol. of CH_3CHO = 0.455 mol

CH_3COOH obtained from 0.313 mol of O_2 = 0.313 2 = 0.626 mol

Thus CH_3CHO is the limiting reactant.

Hence CH_3COOH formed = 0.455 mol

= 0.455 60 = **27.3 g**

(b) O_2 is in excess

0.455 mol of CH_3CHO will react with $\frac{0.455}{2}$ mol of O_2

= 0.2275 mol of O_2

O_2 left unreacted = 0.313 – 0.2275

= 0.0855 mol

= **2.736 g O_2**

(c) Percentage yield = $\frac{\text{actual yield}}{\text{thheoretical yield}}$ 100

$= \frac{23.8}{27.3} \times 100$ = **87.2%**

Example 10

Titanium, which is used to make airplane engines and frames, can be obtained from titanium tetrachloride, which in turn is obtained from titanium oxide by the following process :

$3TiO_2$ (s) + 4C (s) + $6Cl_2$ (g) ⟶ $3TiCl_4$ (g) + $2CO_2$ (g) + 2CO (g)

A vessel contains 4.15 g TiO_2, 5.67 g C and 6.78 g Cl_2 suppose the reaction goes to completion as written, how many grams of $TiCl_4$ can be produced ? (Ti = 48)

Solution :

$$3TiO_2 (s) + 4C (s) + 6Cl_2 (g) \rightarrow 3TiCl_4 (g) + 2CO_2 (g) + 2CO (g)$$

Firt we determine the quantity of $TiCl_4$ from each reactant.

4.15 g TiO_2 = 0.052 mol of TiO_2 = 0.052 mol of $TiCl_4$

5.67 g C = 0.4725 mol of C = 0.3543 mol of $TiCl_4$

6.78 g Cl_2 = 0.0955 mol of Cl_2 = 0.0477 mol of $TiCl_4$

Thus Cl_2 is limiting reactant.

$TiCl_4$ obtained = 0.0477 mol = **9.072 g.**

Example 11

Calculate the weight of FeO produced from 2 g VO and 5.75 g of Fe_2O_3. Also report the limiting reagent.

$$\mathbf{VO + Fe_2O_3 \rightarrow FeO + V_2O_5}$$

Solution Balancing the given equation

	$2VO$ +	$3Fe_2O_3$ →	$6FeO$ +	V_2O_5
Mole before reaction	$\frac{2}{67}$	$\frac{5.75}{160}$		
	= 0.0298	0.0359	0	0
Mole after reaction	$\left(0.0298 - \frac{0.0359 \times 2}{3}\right)$	0	0.0359 2	$0.0359 \times \frac{1}{3}$

∴ Mole of FeO formed = 0.0359 2

∴ Weight of FeO formed = 0.0359 2 72

= 5.17 g

Example 12

A mixture of FeO and Fe_3O_4 when heated in air to a constant weight, gains 5% in its weight. Find the composition of the initial mixture.

Solution :

When FeO and Fe_3O_4 are heated, both change to Fe_2O_3. Let the weights of FeO and Fe_3O_4 be x g and y g respectively.

$\therefore$ total weight of reactant = (x + y) g

Since weight increases 5% on heating,

$\therefore$ when FeO and Fe_3O_4 change completely to Fe_2O_3,

$$\text{the wt. of } Fe_2O_4 = \frac{105}{100} \times (x + y) = 1.05\,(x + y)g$$

Now,

$$\underset{x\ g}{FeO} + \underset{y\ g}{Fe_3O_3} \rightarrow \underset{1.05\ (x + y)\ g,}{Fe_2O_3}$$

Now for Fe atom,

moles of Fe in FeO + moles of Fe in Fe_3O_4 = moles of Fe in Fe_2O_3

1 moles of FeO + 3 moles of Fe_3O_4 = 2 moles of Fe_2O_3

{In this example, since we do not know the balanced equation, but all the products and reactant contain Fe, thus we can directly balance Fe atom and solve it}

$$\frac{x}{72} + \frac{3y}{232} = \frac{2 \times 1.05(x + y)}{160}$$

$$\left\{\begin{array}{c} FeO = 72 \\ Fe_3O_4 = 232 \\ Fe_2O_3 = 160 \end{array}\right\}$$

Dividing by y, we get

$$\frac{1}{72} \times \frac{x}{y} + \frac{3}{232} = \frac{2 \times 1.05}{160} \times \frac{x}{y} + \frac{2 \times 1.05}{160}$$

$$\frac{x}{y}\left(\frac{1}{72} - \frac{2.1}{160}\right) = \frac{2.1}{160} - \frac{3}{232}$$

$$\therefore \quad \% \text{ of FeO} = \frac{81}{(81 + 319)} \times 100 = 20.02\%$$

$$\text{and } \% \text{ of } Fe_3O_4 = \frac{319}{(81 + 319)} \times 100 = 79.98\%.$$

Example 13

Crude CaC_2, is made in an electric furnace by the following reaction :

$$CaO + 3C \rightarrow CaC_2 + CO \uparrow$$

The product contains 85% CaC_2 and 15% unreacted CaO.

(a) How much CaO is to be added to the furnace charge for each 1000 Kg of pure CaC_2 produced ?

(b) How much CaO is to be added to the furnace charge for each 1000 Kg of crude product ?

Solution :

(a) for pure CaC_2 (1000 Kg)

$$1 \text{ mole } CaC_2 \equiv 1 \text{ mol of CaO}$$

$$\Rightarrow \quad 1000/64 \text{ mole} \equiv 1000/64 \text{ mole of CaO}$$

(1000/64) 56 kg of CaO

Also, the product (final) contains 15% unreacted CaO.

for 85 kg of CaO (pure) produced, 15 kg is CaO. (unreacted)

$$1000 \text{ kg of CaO} \equiv \left(\frac{15}{85} \times 1000\right) \text{ kg of unreacted CaO}$$

$$\text{total amount of CaO to be used} = \left(\frac{1000}{64} \times 56\right) + \left(\frac{15}{85} \times 1000\right) \text{kg}$$

$$= 875 + 176.47 = 1051.47 \text{ kg}$$

(b) for crude CaC_2 (1000 kg)

In 1000 kg crude, there will be 150 kg CaO & 850 pure CaC_2.

so amount of CaO to be used

$$= 150 + \frac{850}{64} \times 56$$

$$= 150 + 743.75 = 893.75 \text{ kg.}$$

2.5 VOLUMETRIC ANALYSIS

The volumetric analysis is an analytical method of estimating the concentration of a substance in a solution by adding exactly same number of equivalent of another substance present in a solution of known concentration.

Titration : The process of determination of concentration of a solution of unknown concentration with the help of a solution of known concentration i.e., standard solution, is known as Titration.

Before approaching towards the depth of voumetric analysis one must know the following terms very clearly.

Molarity (M) : The no. of moles of solute present in one litre of solution is called the molarity (M).

$$\text{Molarity} = \frac{\text{No. of moles of solute}}{\text{Volume of solution in litre}}$$

or

$$\text{Molarity} = \frac{\text{Wt. of solute/Molecular wt. of solute}}{\text{Volume of solution in litre}}$$

So, no. of moles of solute = Volume of solution (in litre) Molarity of solution

Molality (m) : No. of moles of solute present in one kilogram of solvent is know as molality.

$$\text{Molality (m)} = \frac{\text{No. of moles of solute}}{\text{Wt. of solvent (in kg)}}$$

Normality (N) : No. of equivalents of solute present in one litre of the solution is known as Normality (N).

$$\text{Normality (N)} = \frac{\text{No. of equivalents of solute}}{\text{Volume of solution in litre}}$$

$$= \frac{\text{Wt. of solute/equivalent weight of solute}}{\text{Volume of solution in litre}}$$

No. of equivalent of solute = Volume of solution (in litre) normality of solution

No. of milli equivalents of solute = Normality of solution Volume of solution in millilitres

Equivalent Weight : Number of parts by mass of an element which reacts or displaces from a compound 1.008 parts by mass of hydrogen, 8 parts by mass of oxygen and 35.5 parts by mass of chlorine, is known as the equivalent weight of that element e.g.

(1)

$$2Mg + O_2 \rightarrow 2MgO$$

48g 32g

12g 8g

$\therefore$ 32g of O_2 reacts with 48g of Mg

$$\therefore \quad 8g \text{ of } O_2 = \frac{48 \times 8}{32} = 12g$$

$\therefore$ Equivalent weight of Mg = 12

Similarly,

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

65.5g 2 1.00g

$\therefore$ 2 1.008 g of hydrogen is displaced by 65.5 g

$$\therefore \quad 1.008 \text{ g of H} = \frac{65.5}{2 \times 1.008} = 32.75g$$

$\therefore$ Equivalent weight of Zn $= \frac{65.5}{2} = 32.75$

(2) $$Al + \frac{3}{2}Cl_2 \longrightarrow AlCl_3$$

32g $\frac{3}{2}$ 71 g $\therefore$ 111.5 g chlorine reacts with 27g of Al

$\therefore$ 35.5 g chlorine reacts with $\frac{27 \times 35.5}{111.5} = 9.0$ g of Al

$\therefore$ Equivalent weight of aluminium $= \frac{27}{3} = 9.0$

As we can see from the above examples that equivalent weight is the ratio of atomic weight and a factor (say n-factor) which is in above three cases is their respective valencies. So,

$$\boxed{\text{equivalent weight} = \frac{\text{atomic weight}}{\text{n} - \text{factor}}}$$

In a similar way, the equivalent weight of acid/base is the ratio of molecular weight and the basicity/acidity and for oxidizing agents and reducing agent it is the ratio of molecular weight and the number of moles of electrons gained or lost. So in case of acid/base the n-factor is basicity/acidity (i.e. number of dissociable H^+ ions/number of dissociable OH^- ion and in case of oxidizing agent/ reducing agent, n-factor is number of moles of electrons gained/lost per mole of oxidizing agent/ reducing agent. Therefore, in general, we can write.

$$\text{Equivalent weight}(E) = \frac{\text{Atomic or molecular weight}}{\text{n} - \text{factor}}$$

$$\text{No. of equivalents of solute} = \frac{\text{Wt}}{\text{Eq. wt.}} = \frac{W}{E} = \frac{W}{M/n}$$

$$\boxed{\text{No. of equivalents of solute} = \text{No. of moles of solute} \quad \text{n-factor}}$$

$$\boxed{\text{And also Normality} = \text{n-factor} \quad \text{molarity of solution}}$$

2.6 n–FACTOR CALCULATION

2.6.1 In Non-Redox Reaction

1. Acids

For acids, n-factor is defined as the number of H^+ ions replaced by 1 mole of acid in a reaction.

Note : That the n-factor for acid is not equal to its basicity; i.e. the number of moles of replaceable H^+ atoms present in one mole of acid.

For example, n–factor of HCl = 1,

n–factor of HNO_3 = 1,

n–factor of H_2SO_4 = 1 or 2, depending upon extent of reaction it undergoes.

$H_2SO_4 + NaOH \rightarrow NaHSO_4 + H_2O$

Although one mole of H_2SO_4 has 2 replaceable H atoms but in this reaction H_2SO_4 has given only one H^+ ion, so its n–factor would be 1.

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$$

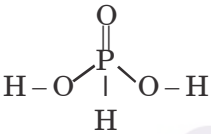
The n–factor of H_2SO_4 in this reaction would be 2.

Similarly, n–factor of H_2SO_3 = 1 or 2

n–factor of H_2CO_3 = 1 or 2

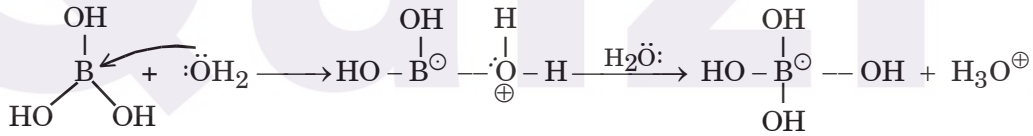
n–factor of H_3PO_4 = 1 or 2 or 3

n–factor of H_3PO_3 = 1 or 2 because one of the H is not replaceable in H_3PO_3.

This can be seen using its structure  . The H atoms which are linked to oxygen are replaceable while the H atom linked directly to central atom (P) is non replaceable.

n–factor of H_3BO_3 = 1

In H_3BO_3, although all three H are linked to oxygen, yet all 3 H are not replaceable. Here boron atom is electron deficient, so its acts as a Lewis acid. When H_3BO_3 oxygen atom of H_2O through its lone pair attack the boron atom, as follows



The net reaction is $H_3BO_3 + 2H_2O \rightarrow [B(OH)_4]^- + H_3O^+$

Thus, one mole of H_3BO_3 in solution gives only one mole of H^+, so its n–factor is 1.

2. Bases

For bases, n–factor is defined as the number of OH^- ions replaced by 1 mole of base in a reaction. Note that n–factor is not equal to its acidity i.e. the number of moles of replaceable OH^- ions present in 1 mole of base.

For example, n–factor of NaOH = 1

n–factor of $Ca(OH)_2$ = 1 or 2

n–factor of $Al(OH)_3$ = 1 or 2 or 3

n–factor of $NH_4(OH)$ = 1

2.6.2 In Redox Reactions

1. Salts in which no atom undergoes change in oxidation state in reactions

The n–factor for such salts is defined as the total moles of cationic/anionic charge replaced in 1 mole of the salt. For the reaction,

$$Na_3PO_4 + BaCl_2 \rightarrow NaCl + Ba_3(PO_4)_2$$

To get one mole of $Ba_3(PO_4)_2$, two moles of Na_3PO_4 are required, which means six moles of Na^+ are completely replaced by 3 moles of Ba^{2+} ions. So, six moles of cationic charge is replaced by 2 moles of Na_3PO_4, thus each mole of Na_3PO_4 replaces 3 moles of cationic charge. Hence, n–factor of Na_3PO_4 in this reaction is 3.

2. Salts which react in a manner that only one atom undergoes change in oxidation state and goes in only one product

The n–factor of such salts is defined as the number of moles of electrons exchanged (lost or gained) by one mole of the salt.

Let us have a salt A_aB_b in which oxidation state of A is + x. It changes to a compound, which has atom D in it. The oxidation state of A in A_cD be + y.

$$A_a^{+x}B_b \longrightarrow A_c^{+y}D$$

The n–factor of A_aB_b is calculated as

$$n = |ax - ay|$$

To calculate the n–factor of such salts, we apply following method :

1. Take one mole of rectant whose oxidation state is changing.
2. Multiply this with oxidation state of element in reactant to give us total oxidation state of element in reactant.
3. Calculate total oxidation state of the same element in the product for same number of moles of atom in reactant.

Another similar case would be that the salt has only one atom undergoing change in oxidation state but going into two different products with the same oxidation state. Try it yourself.

For example, let us calculate the n–factor $KMnO_4$ for the given chemical change.

$$KMnO_4 \xrightarrow{H^+} Mn^{+2}$$

In this reaction, oxidation state of Mn changes from + 7 to + 2. Thus; $KMnO_4$ is acting as oxidising agent, since it is reduced.

$\therefore$ n–factor of $KMnO_4 = |1 \quad (+7) - 1 \quad (+2) = 5$

Similarly

(a) $KMnO_4 \xrightarrow{H_2O} Mn^{+4}$

n–factor of $KMnO_4 = |1 \quad (+7) - 1 \quad (+4) = 3$

(b) $KMnO_4 \xrightarrow{OH^-} Mn^{+6}$

n–factor of $KMnO_4$ = |1 (+ 7) – 1 (+ 6)| = 1

It can be seen that in all the above chemical changes, $KMnO_4$ is acting as oxidising agent, yet its n–factor is not same in all reactions. Thus, the n–factor of a compound is not fixed, it depends on the type and the extent of reaction it undergoes.

Example 14

Calculate the n–factor of reactants in the given chemical changes ?

(a) $\mathbf{K_2Cr_2O_7 \xrightarrow{H^+} Cr^{3+}}$

(b) $\mathbf{C_2O_4^{2-} \longrightarrow CO_2}$

(c) $\mathbf{S_2O_3^{2-} \xrightarrow{alkaline} SO_4^{2-}}$

(d) $\mathbf{I^- \longrightarrow ICl}$

Solution :

(a) $K_2 \overset{+6}{Cr_2} O_7 \xrightarrow{H^+} Cr^{3+}$

The oxidation state of Cr changes from + 6 to + 3.

∴ n = |2 (+ 6) – 2 (+ 3)| = 6

(b) $\overset{+3}{C_2} O_4^{2-} \longrightarrow \overset{+4}{C}O_2$

Carbon get oxidized from + 3 to + 4

∴ n = |2 (+ 3) – 2 (+ 4)| = 2

(c) $\overset{+2}{S_2} O_3^{2-} \xrightarrow{alkaline} \overset{+6}{S} O_4^{2-}$

The oxidation state of sulphur changes from + 2 to + 6

∴ n = |2 (+ 2) – 2 (+ 6)| = 8

(d) $I^- \longrightarrow \overset{+1}{I}Cl$

I^- get oxidized to I^+.

∴ n = |1 (– 1) (+ 1)| = 2

3. Salts which react in a manner that only one atom undergoes change in oxidation state but goes in two products with different oxidation state (different than in the reactant) as a result of either oxidation or reduction.

Let the chemical change be

$$A_a^{+x}B_b \longrightarrow A_c^{+y}D + A_e^{+z}F$$

The number of moles of electrons lost or gained by one mole of A_aB_b would depend on the fact that how much of A underwent change to oxidation state + y and how much of A underwent change to oxidation state + z. n–factor calculation in such cases can be done as follows.

Let us take a chemical change $2Mn^{+7} \longrightarrow Mn^{+4} + Mn^{+2}$ out of the two moles of Mn^{+7}, one mole Mn^{+7} changes to Mn^{+4} by gaining 3 mole of electrons and the other mole of Mn^{+7} changes to Mn^{+2} by gaining 5 mole of electrons, so in all 8 mole of electrons are gained by 2 mole of In^{+7}. So each mole of Mn^{+7} has gained 8/2 = 4 mole of electrons. Thus, 4 would be the n–factor on Mn^{+7} in this reaction.

If the reaction would have been

$$3Mn^{+7} \longrightarrow 2Mn^{+2} + Mn^{+4}$$

Out of 3 moles of Mn^{+7}, two moles of Mn^{+7} changes to Mn^{+2} by gaining 10 mole of electrons and one mole of Mn^{+7} changes to Mn^{+4} by gained 3 mole of electrons. Thus each mole of Mn^{+7} have gained 13/3 mole of electrons. Therefore, the n–factor of Mn^{+7} in this reaction would be 13/3.

"Note that n–factor can be a fraction because it is not the number of electrons exchanged but it is the number of moles of electrons exchanged which can be a fraction."

4. Salts that react in a manner that two type of atoms in the salt undergo change in oxidation state (both the atoms are either getting oxidised or reduced).

Let the change be represented as

$$A_a^{+x}B_b \longrightarrow A_c^{+y}D + E_fB^{+z}$$

In such cases, the n–factor of the compound would be the sum of individual n–factors of A and B.

n–factor of A = $|ax - ay|$

n–factor of B = $|ax - bz|$ because the total oxidation state of 'b' B's in the reactant is – ax (as the total oxidation state of 'a' A's in the reactant is + ax and the total oxidation state of y B's in the product is bz.

$\therefore$ n–factor of A_aB_b = $|ax - ay| + |ax - bz|$

In general, the n–factor of the salt will be the total number of mole of electrons lost or gained by one mole of the salt.

For example, we have a reaction, $Cu_2^{+1}S^{-2} \longrightarrow Cu^{2+} + \overset{+4}{S}O_2$

in which Cu^+ and S^{2-} both are getting oxidised to Cu^{2+} and S^{+4} respectively.

$\therefore$ n–factor of $Cu_2S = \left|2 \quad (+1) - 2 \quad (+2)\right| + \left|1 \quad (-2) - 1 \quad (+4)\right| = 8$

5. Salts that react in a manner that two atoms in the salt undergoes change in oxidation state (one atom is getting oxidised and the other is getting reduced).

The n–factor of such a salt can be calculated either by taking the total number of moles of electrons lost or total number of mole of electrons gained by one mole of the salt.

$$K\overset{+5}{Cl}\overset{-2}{O_3} \longrightarrow K\overset{-1}{Cl} + \overset{0}{O_2}$$

In this reaction, O^{2-} is getting oxidised to O_2 and Cl^{+5} is getting reduced to Cl^{-1}. In each case, 6 mole of electrons are exchanged whether we consider oxidation or reduction.

n–factor of $KClO_3$ considering oxidation = $|3(-2) - 3(0)| = 6$

or n–factor of $KClO_3$ considering reduction = $|1 \quad (+5) - 1 \quad (-1) = 6$

6. Salts or compounds which undergoes disproportionation reaction

Disproportionation reactions are the reactions in which oxidising and reducing agents are same or the same element from the same compound is getting oxidised as well as reduced. n–factor of a disproportionation reaction can only be calculated using a balanced chemical reaction. We will categorize disproportionation reactions into two types.

(a) Disproportionation reactions in which moles of reducing and oxidizing agents are the same.

In such compounds we either calculate the n–factor for reducing agent or oxidizing agent since calculating both would be simply a waste of time.

For example, $2H_2O_2 \rightarrow 2H_2O + O_2$

Out of the 2 mole of H_2O_2 used in reaction, one mole of H_2O_2 gets oxidised to O_2 (oxidation state of O changes from – 1 to 0) while the other mole of H_2O_2 gets reduced to H_2O (oxidation state of O changes from – 1 to – 2). When 1 mole of H_2O_2 gets oxidised to O_2, the half-reaction would be $O_2^{2-} \longrightarrow O_2^0 + 2e^-$ and when 1 mole of H_2O_2 gets reduced to H_2O, the half-reaction would be $O_2^{2-} + 2e^- \longrightarrow 2O^{2-}$

Thus, it is evident that one mole of H_2O_2 (which is either getting oxidised or reduced) will lose or gain 2 mole of electrons. Therefore n–factor of H_2O_2 as oxidizing as well as reducing agent in this reaction is 2. Thus

$$H_2O_2 + H_2O_2 \rightarrow 2H_2O + O_2$$

Reducing agent (n = 2) Oxidizing agent (n = 2) (n = 1) (n = 2)

(b) Disproportionation reactions in which moles of compound getting oxidised and reduced are not same i.e. moles of oxidising agent agent and reducing agent are not same.

For example,

$$6Br_2 + 12\ OH^- \rightarrow 10Br^- + 2BrO_3^- + 6H_2O$$

In this reaction, the mole of electrons lost by the oxidation of some of the moles of Br_2 are same as the number of mole of electrons gained by the reduction of rest of the moles of Br_2. Of the 6 moles of Br_2 used, one mole is getting oxidized, loosing 10 electrons (as reducing agent) and 5 moles of Br_2 are getting reduced and accepts 10 moles of electron (as oxidizing agent).

$$Br_2 \longrightarrow 2Br^{+5} + 10e^-$$

$$5Br_2 + 10e^- \longrightarrow 10Br^-$$

$$Br_2 + 5Br_2 \longrightarrow 10Br^- + 2Br^{+5}$$

Reducing agent Oxidizing agent

(n = 10) (n = 2) (n = 1) (n = 5)

Thus, n–factor of Br_2 acting as oxidizing agent is 2 and that Br_2 acting as reducing agent has n–factor 10.

Or when the reaction is written as

$$6Br_2 \rightarrow 10Br^- + 2Br^{+5}$$

where, Br_2 is not distinguished as how much of it functions as oxidizing agent and how much as reducing agent, then for calculating n–factor of compound in such reactions, first find the total number of mole of electrons exchanged (lost or gained) using the balanced equation and divide it with the number of mole of Br_2 involved in the reaction to get the number of mole of electrons exchanged by one mole of Br_2.

In the overall reaction, the number of mole of electrons exchanged (lost or gained) is 10 and the moles of Br_2 used in the reaction are 6. Thus, each mole of Br_2 has exchanged 10/6 or 5/3 mole of electrons. Therefore, the n–factor of Br_2 when the reaction is written without segregating oxidising and reducing agent is 5/3.

$$\underset{(n = 5/3)}{6Br_2} \rightarrow \underset{(n = 1)}{10Br^-} + \underset{(n = 5)}{2Br^{+5}}$$

2.10 LAW OF EQUIVALANCE AND DILUTION EFFECT

Law of Equivalence

It is not practically possible to remember all the chemical reactions in balanced forms though that is how we require them to be for solving problems.

Hence we use the law of equivalence. This law helps us determine the molar ratio of reactants and products. It can be stated as :

According to the law of equivalence, whenever two substances react, the equivalents of one will be equal to the equivalents of other and the equivalents of any product will also be equal to that of the reactant.

Law of equivalence

For a reaction

$$aA + bB \rightarrow mM + nN$$

eq. of A = eq. of B = eq. of M = eq. of N

or $$\frac{\text{wt. of A}}{E_A} = \frac{\text{wt. of B}}{E_B} = \frac{\text{wt. of M}}{E_M} = \frac{\text{wt. of N}}{E_N}$$

Let us suppose we have a reaction, A + B → C + D. In this reaction, the number of moles of electrons lost by 1 mole of A are x and the number of mole of electrons gained by 1 mole of B are y. Since, the number of mole of electrons lost and gained are not same, the molar ratio in which A & B react cannot be 1 : 1. Thus, if we take y moles of A, then the total moles of electrons lost by y moles of A would be (x y). Similarly, if x moles of B are taken, then the total mole of electrons gained by x moles of B would be (y x). Thus, the number of electrons lost by A and number of electrons gained by B becomes equal. For reactant A, its n–factor is x and the number of moles used are y. So,

The equivalents of A reacting = moles of A reacting x n–factor of A = y x

Similarly, for reactant B, its n–factor is y and the number of moles used are x. So,

The equivalents of B reacting = moles of B reacting n–factor of B

= x y

Thus, the equivalents of A reacting would be equal to the equivalents of B reacting. Thus, the balancing coefficients of the reactant would be as

yA	+	xB	→	C	+	D
(n–factor = x)		(n–factor = y)				

The n–factor of A & B are in the ratio of x : y, and their molar ratio is y : x. Thus, molar ratio is inverse of the n–factor ratio.

In general, whenever two substances react with their n–factors in the ratio of a : b, then their molar ratio in a balanced chemical reaction would be b : a.

To get the equivalents of a substance, its n–factor is to be known. Let the weight of the substance used in the reaction be w g.

Then, equivalents of substance reacted would be $\frac{w}{E}$ or $\frac{w}{M_1/n} = \frac{w}{M_1} \times n$ (where E and M_1 are the equivalent mass and molar mass of the substance) Thus, in order to calculate the equivalents of a substance, knowledge of n–factor is a must.

When a solution is diluted, the moles and equivalents of solute do not change but molarity and normality change while on taking out a small volume of solution from a larger volume, the molarity and normality of solution do not change but molar and equivalents change proportionately. This concept is useful in solving dilution problems.

2.11 Some Important Formulae

1. moles = $MV_{lt} = \frac{g_B}{M_B}$ (M = Molarity of solution

V = Volume of solution_

millimoles = $MV_{cc} = \frac{g_B}{M_B} \times 100$ g_B/M_B = Gram equivalent of substance

2. gm eq. = $NV_{lt} = \frac{g_B}{E_B}$ N = Normality of solution

V = volume

meq = $NV_{cc} = \frac{g_B}{E_B} \times 1000$ $\frac{g_B}{E_B}$ = Gram equivalents

3. Molarity (M) = $\frac{10^3 \times d}{M_0}$ Normality (N) = $\frac{10^3 \times d}{E}$

d : density of solution in g/cc; x : %age strength of solution (by weight)

M_0 = Molar mass of substance

4. N = x M

x : acidity for base or basicity for acid or electron transfer/mole for O.A and R.A

N = Normality

M = Molarity

5. $m = \frac{1000 \times x_B}{M_A \times x_A}$

m = molarity

x_B = mole fraction of B

x_A = mole fraction of A

6. **Strength of a solution** is generally expressed as grams of solute in one litre of solution.

Strength = $\frac{g_B}{\text{litres of solution}}$

Strength = NE = M M_0

Example 15

How many milli equivalent of H_2SO_4 in 10 ml. of $\frac{N}{10}$ H_2SO_4.

Solution :

$$\text{no. of equivalent} = NV \text{ (in lt)}$$

$$\text{no. of m.eq.} = NV \quad (100)$$

$$= \frac{1}{10} \times \left(10 \times 10^{-3}\right) \times 1000$$

$$\text{m.eq.} = 1$$

Example 16

What volume of 0.1 M $KMnO_4$ in acidic medium will be required to react completely with 100 ml; 0.1 M $NaHC_2O_4$.

Solution :

In redox reaction, n-factor of $NaHC_2O_4$ is 2.8 and that of $KMnO_4$ is 5.

Therefore,

From law of equivalence,

$$(N_1\ V_1)\ KMnO_4 = (N_2\ V_2) NaHC_2O_4$$

$$0.1 \quad 5 \quad V_1 = 0.1 \quad 2 \quad 100 \Rightarrow V_1 = 40 \text{ ml}$$

Example 17

What volume of 0.1 NaOH will be required in previous problem.

Solution :

NaOH + $NaHC_2O_4$ Acid base reaction

Here, n-factor of $NaHC_2O_4$ = 1

$$(N_1\ V_2)\ NaHC_2O_4 = (N_2\ V_2)\ NaOH \text{ [from law of equivalence]}$$

$$\therefore \quad 100 \quad 0.1 = .1 \quad V$$

$$V_2 = 100 \text{ ml}$$

N_1 = Normality of $NaHC_2O_4$

N_2 = Normality of NaOH

V_1 = Volume of $NaHC_2O_4$

V_2 = Volume of NaOH

Example 18

What is the normality and nature of a mixture obtained by mixing 0.62 g of $Na_2C_2O_3$. H_2O to 100 mL of 0.1 N H_2SO_4 ?

Solution :

$$\text{Meq. of } Na_2CO_3.H_2O = \frac{0.62}{62} \times 100 \left(\frac{w}{E} \times 1000 = \text{Meq.}\right)$$

Meq. of H_2SO_4 = 100 0.1

	Na_2CO_3	+	H_2SO_4	$\rightarrow$	Na_2SO_4	+	H_2O	+	$CO_2\uparrow$
Meq. added	10		10		0		0		0
Meq. left	0		0		10		10		10

$$\therefore \quad N_{Na_2SO_4} = \frac{10}{100} = 0.1$$

Solution becomes neutral since both acid and base are used up and Na_2SO_4 does not show hydrolysis.

Example 19

A sample of an alloy weighing 0.50 g and containing 90% Ag was dissolved in conc. HNO_3. Ag was analysed by Volhard method in which 25 mL of KCNS were required for complete neutralization. Determine normality of KCNS.

Solution :

$$\text{Wt. of Ag} = \frac{90 \times 0.5}{100} = 0.45$$

$$\therefore \quad \text{Meq. of Ag} = \frac{0.45}{108} \times 1000 = 4.17$$

$$\text{NowAg} \xrightarrow{HNO_3} AgNO_3 \xrightarrow{KCNS} AgCNS$$

$\therefore$ Equal Meq. reacts and therefore

$$\text{Meq. of KCNS} = \text{Meq. of } AgNO_3$$

$$= \text{Meq. of } HNO_3 = \text{Meq. of Ag}$$

$$N \quad 25 = 4.17$$

$$N = 0.167$$

Example 20

HNO_3 used as a reagent has specific gravity of 1.42 g mL^{-1} and contains 70% by strength HNO_3. Calculate,

(a) Normality of acid

(b) Volume of acid that contains 63 g pure acid

(c) Volume of water required to make 1N solution from 2 mL conc. HNO_3.

Solution : (a) Strength of HNO_3 = 70%

i.e. Volume of solution = 100 mL

and Volume of HNO_3 = 70 mL

$\therefore$ Wt of HNO_3 in solution = 70 1.42 g

$$\therefore \quad \text{Eq. of } HNO_3 \text{ in solution} = \frac{70 \times 1.42}{63 \times 100/1000} \left(\because N = \frac{\text{Eq.}}{\text{V in litre}} \right)$$

$$= \mathbf{15.78}$$

(b) $\therefore$ 70 1.42 g of pure acid is present in 100 mL

$$\therefore \quad 63 \text{ g of pure acid present in} = \frac{100 \times 63}{70 \times 1.42}$$

$$= \mathbf{63.38\ mL}$$

(c) Meq. of conc. HNO_3 = Meq. of dil. HNO_3

$\therefore$ 2 15.78 = V 1

V = 31.56 mL

$\therefore$ Volume of water added = 31.56 – 2 = **29.56 mL**

2.12 Volumetric Analysis

Now, we have developed enough platforms to understand the law of equivalents and volumetric before analysis.

The volumetric analysis is an analytical method of estimating the concentration of a substance in a solution by adding exactly same number of equivalents of another substance present in a solution of known concentration.

This is the basic principle of titration. Volumetric analysis is also known as **titrimetric analysis.**

The substance whose solution is employed to estimate the concentration of unknown solution is called **titrant** and the substance whose concentration is to be estimated is called **titrate.**

The volumetric analysis is divided into following types :

(a) Simple titrations

(b) Back titrations

(c) Double titrations

SIMPLE TITRATION

A known volume of the solution of unknown concentration is taken in a flask and required reagents are added to it. The solution of known concentration is added from the burette in the solution of unknown concentration till the latter reacts completely. This process is called titration. At the end point (equivalence point) the equivalents or milliequivalents of the two reacting substances are equal.

$$\text{Volume of solution (A)} = V_A \text{ litres}$$

$$\text{Normality of solution (A)} = N_A$$

$$\text{Equivalents of substance (A)} = N_A\ V_A$$

$$\text{Similarly, equivalents of substance (B)} = N_B\ V_B$$

At the equivalence point (end point) the equivalents (not the moles) of the two substances equal.

$$N_A\ V_A \text{ (litre)} = N_B \quad V_B \text{ (litre)}$$

Example 21

40 ml N/10 HCl and 60 ml N/20 KOH are mixed together. Calculate the normality of the acid or base left. What is the normality of the salt formed in the solution?

Solution :

$$\text{Milli equivalents of HCl} = N \quad V\text{ (ml)} = \frac{1 \times 40}{10} = 4$$

$$\text{Milli equivalents of KOH} = N \quad V\text{ (ml)} = \frac{1 \times 60}{20} = 3$$

One milli equivalent of an acid neutralizes one milli equivalent of a base

$$\text{Milli equivalent of HCl left} = 4 - 3 = 1$$

$$\text{Total volume of the solution} = 40 + 60 = 100 \text{ ml}$$

$$\text{Milli equivalents of HCl} = N \quad V\text{ (ml)}$$

$$1 = N \quad 100$$

$$\text{Normality (N) of HCl left in solution} = 0.01$$

Salt formed = Milli equivalent of acid or base neutralized

$$\text{Milli equivalents of the salt formed} = N \quad V\text{ (ml)}$$

$$3 = N \quad 100$$

$$\text{Normality (N) of salt formed} = 0.03$$

DOUBLE TITRATION

If an aqueous solution contains a mixture of any two of the three NaOH, $NaHCO_3$ and Na_2CO_3 and it has to be titrated against an acid HCl or H_2SO_4, it will require two indicators to determine the strength of the bases present. The two indicators used are phenophthalein and methyl orange.

Method

A given volume of the aqueous solution of the bases is taken and phenophthalein indicator is added to it. This solution is titrated with an acid of known normality to the end point the volume of the acid used to noted. This end point is called first end point. Now in the same solution methyl orange is added and gain titrated with an acid of known strength to the end point. It is called second end point. The volume of acid, used in the second end point is also noted. The normality of the bases present is then calculated.

Principle : In the presence of phenophthalein indicator

(i) NaOH is completely neutralized by the acid

(ii) Only half of the milliequivalents of Na_2CO_3 present are titrated as the phenophthalein will show the colour change when only $NaHCO_3$ (weak base) and neutral species are left in the reaction mixture. The following reactions take place,

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

$$Na_2CO_3 + HCl \longrightarrow \underset{\text{weak base}}{NaHCO_3} + NaCl$$

Since phenolphthalein is a weak organic acid, and it changes its colour in weakly basic medium ($NaHCO_3$), so as soon as the Na_2CO_3 is converted to $NaHCO_3$, phenophthalein shows the colour change indicating the completion of the reaction.

In the presence of methyl orange, all the basic structures, left in the mixture will be neutralized by acid and methyl orange will show the colour change when the medium is weakly acidic ($H_2O + CO_2$ i.e. H_2CO_3).

1. **Titration of the solution containing both NaOH and Na_2CO_3 :** A given volume of the aqueous solution is titrated with an acid of known normality using phenophthalein indicator. Suppose 'a' milli equivalents of acid are used in the first end point then,

   milli equivalents of NaOH + ½ milli equivaent of Na_2CO_3

   = milli equivalent of acid = a .... (1)

   Now in the same already titrated solution methyl orange indicator is added and again titrated to the end point. Suppose 'b' milli equivalents of the acid are used at the second end point.

   ½ milli equivalents of Na_2CO_3 = milli equivalents of acid = b ..... (2)

From equation (1) and (2)

Milli equivalents of acid used by Na_2CO_3 = 2b

= milli equivalents of Na_2CO_3

Milli equivalents of acid used by NaOH = a – b

= milli equivalent of NaOH

Knowing the milli equivalents of Na_2CO_3 or NaOH and the volume of the solution titrated, their normality can be calculated.

Example 22

NaOH and Na_2CO_3 are dissolved in 200 ml aqueous solution. In the presence of phenophthalein indicator, 17.5 ml of 0.1 N HCl are used to titrate this solution. Now methyl orange is added in the same solution titrated and it requires 2.5 ml of the same HCl. Calculate the normality of NaOH and Na_2CO_3 and their mass present in the solution.

Solution :

Milli equivalents (a) of HCl used in the presence of phenolphathalein indicator

= N V (ml)

= 0.1 17.5 = 1.75

1.75 (a) = milli eq. of NaOH + ½ milli eq. of Na_2CO_3 .... (1)

Milli eq. (b) of HCl used in the presence of methyl orange indicator

= N V (ml)

= 0.1 2.5 = 0.25

0.25 (b) = ½ milli equivalents of Na_2CO_3 ... (2)

For Na_2CO_3 solution : From equation (2)

Milli eq. of acid used by Na_2CO_3 = 2b

= 2 0.25 = 0.5

Volume of Na_2CO_3 solution = 200 ml

Suppose, Normality of Na_2CO_3 = N

Milli equivalents of Na_2CO_3 = N V (ml) = 200 N

Putting equivalents of acid and Na_2CO_3 equal

200N = 0.5

or (Normality of Na_2CO_3 solution) N = $\frac{1}{400}$

Mass of Na_2CO_3 = N E V (litre)

(E for Na_2CO_3 = 53) $= \frac{1}{400} \times 53 \times 0.2$

= 0.0265 gram

For NaOH solution : From equation (1) and (2)

Milli eq. acid used by NaOH = a – b = 1.75 – 0.25

= 1.50

Volume of NaOH solution = 200 ml

Suppose, Normality of NaOH solution = N

Milli eq. of NaOH = N V (ml) = 200 N

Putting the milli eq. of NaOH and acid used equal

200 N = 1.5

(Normality of NaOH solution) $N = \frac{1.5}{200}$

Mass of NaOH = N E V (litres)

$= \frac{1.5}{200} \times 40 \times 0.2$

(E for NaOH = 40) = **0.06 g**

2. **Titration of the solution containing both Na_2CO_3 and $NaHCO_3$:** Given volume of the solution is titrated by an acid using phenolphthalein indicator. Suppose 'a' milli equivalents of acid are used in the first end point. Then ½ milli equivalent of Na_2CO_3 = milli equivalents of acid ... (1)

Now in the same already titrated solution methyl orange indicator is added and again titrated to the end point : Suppose 'b' milli equivalents of the acid are used at the second end point. Then

½ milli equivalents of Na_2CO_3 + milli equivalents of $NaHCO_3$

= milli equivalents of acid = b ... (2)

From equation (1) and (2)

Milli equivalents of acid used by Na_2CO_3 = 2a

= milli equivalents of Na_2CO_3

Milli equivalents of acid used by $NaHCO_3$ = b –a

= milli equivalent of $NaHCO_3$

Knowing the milli equivalents of the base and volume of the solution titrated, the normality (strength) of the bases can be calculated.

Example 23

Both Na_2CO_3 and $NaHCO_3$ are present in an aqueous solution. In the presence of phenolphthalein indicator 10 ml of this solution requires 2.5 ml of 0.1 M H_2SO_4 for titration. After this methyl ornge is added in the same solution and titration requires 5 ml M H_2SO_4. Calculate the concentration of Na_2CO_3 and $NaHCO_3$ in g/litre.

Solution :

Milli eq. (a) of H_2SO_4 used in the presence of phenolphthalein indicator

$= N \quad V\,(ml) = 0.1 \quad 2 \quad 2.5 = 0.5$

$a = 0.5 = \frac{1}{2}$ milli equivalents of Na_2CO_3 ... (1)

Milli eq. of $NaHCO_3$ + ½ milli eq. of Na_2CO_3 = (b) ... (2)

For Na_2CO_3 solution : From equation (1)

Milli eq. of acid used by Na_2CO_3 = 2 0.5 = 1

Suppose, Normality of Na_2CO_3 solution = N

Volume of Na_2CO_3 solution taken = 10 ml

Putting the milli eq. of H_2SO_4 and Na_2CO_3 equal

1 = 10 N

or (Normality of Na_2CO_3) N = 0.1

Strength (S) in g/litre = N E

= 0.1 53 (E for Na_2CO_3 = 53)

= 5.3 g/litre

For $NaHCO_3$ solution : From equation (1) and (2) milli eq. of acid used by

$NaHCO_3$ = b – a = 1.0 – 0.5 = 0.5

Suppose, Normality of $NaHCO_3$ solution = N

Volume of $NaHCO_3$ solution taken = 10 ml

Milli equivalents of $NaHCO_3$ taken = 10 N

Putting the milli eq. of H_2SO_4 and $NaHCO_3$ equal,

0.5 = 10 N

or (Normality of $NaHCO_3$ solution) N = 0.05

Strength (S) in g/litre = N E

(E for $NaHCO_3$ = 84) = 0.05 84 = **4.2 g/litre**

BACK TITRATION

Let us consider that we have an impure solid substance 'Z' weighing 'w' g and we are required to calculate the percentage purity of 'Z' in the sample. We are also provided with two solutions 'X' and 'Y', where the concentration of 'Y' is known (N_1) and that of 'X' is unknown. For the back titration to work, following conditions are to be satisfied.

(a) Compounds 'X', 'Y' and 'Z' should be such that 'X' and 'Y' reacts with each other.

(b) 'X' and pure 'Z' also reacts with each other but impurity present in 'Z' does not react with 'Z'.

$$Z + X \text{ (excess)} \rightarrow \text{Product 1}$$

$$\text{Remaining (X)} + Y \rightarrow \text{Product 2}$$

Note : Product 1 should not react with Y

Milli equivalent of Y = N_2V_2

Where N_2 and V_2 (ml) is the normality and volume of Y

Initial milli equivalent of X = N_1V_1

Where N_1 and V_1 (ml) is the normality and volume of X

Remaining milli equivalents of X after reacting with Y = $N_1V_1 - N_2V_2$

Remaining milli equivalents of X = milli equivalents of Z

$$N_1V_1 - N_2V_2 = \frac{a \quad 100}{\text{Equivalent weight}}$$

Where 'a' is the weight of pure Z which is reacted.

$$a = \frac{\text{Molecular weight}\left(N_1V_1 - N_2V_2\right)}{n - \text{factor}}$$

$$\therefore \quad \text{Percentage purity of 'Z'} = \frac{\left(N_1V_1 - N_1V_2\right)}{n - \text{factor}} \times \frac{\text{Molar Mass of Z}}{W} \times 100$$

Example 24

20 g of sample of $Ba(OH)_2$ is dissolved in 10 ml of 0.5 N HCl solution. The excess of HCl was titrated with 0.2 N NaOH. The volume of NaOH used was 10 cc. Calculate the percentage of $Ba(OH)_2$ in the sample.

Solution :

Milli eq. of HCl initially= 10 0.5 = 5

Milli eq. of NaOH consumed = Milli eq. of HCl in excess

= 10 0.2 = 2

$\therefore$ Milli eq. of HCl consumed = Milli eq. of $Ba(OH)_2$

= 5 – 2 = 3

$\therefore$ eq. of $Ba(OH)_2$ = 3/1000 = 3 10^{-3}

Mass of $Ba(OH)_2$ = 3 10^{-3} (171/2) = 0.2565 g

% of $Ba(OH)_2$ = (0.2565/20) 100 = 1.28%

(i) IODIMETRY

This titration involves free iodine. Such direct estimation of iodine is called iodimetry. This involves the titration of iodine solution with known sodium thiosulphate solution, whose normality is N. Let the volume of sodium thiosulphate used by V litre.

$$\underset{(n=2)}{I_2} + \underset{(n=1)}{2Na_2S_2O_3} \longrightarrow 2NaI + Na_2S_4O_6$$

$$\text{Equivalents of } I_2 = \text{Equivalents of } Na_2S_2O_3 \text{ used} = N \quad V$$

$$\text{Moles of } I_2 = \frac{N \times V}{2}$$

$$\text{Mass of free } I_2 \text{ in the solution} = \left[\frac{N \times V}{2} \times 254\right] g$$

(ii) IODOMETRY

This is an indirect method of estimation of iodine. An oxidising agent is made to react with excess of solid KI. The oxidising agent oxidises I^- to I_2. This liberated iodine is then made to react with $Na_2S_2O_3$ solution of normality N. Let the volume of thiosulphate solution required be V litre.

$$\text{Oxidising Agent (A)} + KI \longrightarrow I_2 \xrightarrow{2Na_2S_2O_3} 2NaI + Na_2S_4O_6$$

reduced form of oxidizing agent

$$\text{Equivalents of 'A'} = \text{equivalents of } I_2 = \text{Equivalents of } Na_2S_2O_3 \text{ used} = N \quad V$$

$$\text{Equivalents of } I_2 \text{ liberated from KI} = N \quad V$$

$$\text{Equivalents of 'A'} = N \quad V$$

Let the n–factor of 'A' in its reaction with KI be x, then

$$\text{Mass of 'A' consumed} = \frac{N \times V}{x} \times M_A \quad (\text{where } M_A \text{ is the molar mass of A})$$

Example 25

A 1.1 g sample of copper are is dissolved and Cu^{2+} (aq.) is treated with KI. I_2 liberated required 12.12 mL of 0.1 M $Na_2S_2O_2$ solution for nitration. What is % of Cu in the ore ?

$$2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$$

$$I_2 + 2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$$

$$2S_2O_3^{2-} \equiv I_2 \equiv 2Cu^{2+}$$

$$\text{Thus } S_2O_3^{2-} \equiv I \equiv Cu^{2+}$$

$$\text{hence equivalent weight of } S_2O_3^{2-} = \text{mol. wt.}$$

$$\text{and that of } Cu^{2+} = \text{mol. wt.}$$

$$\text{Equialent of } Na_2S_2O_3 = \text{equivalent of } I_2$$
$$= \text{equivalent of } Cu^{2+}$$
$$w\ (\text{pure (Cu)}) = \frac{NEV}{1000}$$
$$= \frac{0.1 \times 63.5 \times 12.12}{1000}$$
$$= 0.077\ g$$

This is present in 1.1 g copper ore.

$\therefore$ % of copper = 7.0%

Example 26

A 0.589 g sample of pyrolusite ore (impure MnO_2) is treated with 1.65 g hydrate oxalic acid ($H_2C_2O_4.2H_2O$) in an acidic medium. Following this reaction, the excess oxalic acid required 30.6 mL of 0.1 M $KMnO_4$. What is % of MnO_2 in pyrolusite ?

Solution : Oxalic acid will react with pure MnO_2

unreacted oxalic acid $\equiv$ 30.06 mL of 0.1 M $KMnO_4$

$$\text{oxalic acid taken} = 1.65\ g = \frac{1.65}{126} = 0.013\,\text{mol}$$

$$MnO_2 + C_2O_4^{2-} + 4H^+ \rightarrow 2CO_2 + Mn^{2+} + 2H_2O \quad ...(i)$$

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O \quad ...(ii)$$

unreacted

30.06 mL of 0.1 M $KMnO_4$ = 30.06 mL of 0.5 N $KMnO_4$.

$$\text{unreacted oxalic acid} = \frac{NEV}{1000}$$
$$= \frac{0.5 \times 63 \times 30.06}{1000}$$
$$= 0.95\ g = \frac{0.95}{126}$$
$$= 0.0075$$

used oxalic acid in reaction (i) = 0.013 – 0.0075

= 0.00558 mol

= 5.58 10^{-3} mol

hence mol of MnO_2 (pure) = 5.58 10^{-3} mol

= 0.485 g

$$\%\ \text{of pure } MnO_2 = \frac{0.485}{0.589} \times 100 = \mathbf{82.42\%}$$

Example 27

A 0.608 g sample of fertiliser contained nitrogen as ammonium sulphate, $(NH_4)_2SO_4$. It was analysed for nitrogen by heating with sodium hydroxide.

$$(NH_4)_2SO_4(s) + 3NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l) + 2NH_3(g)$$

The ammonia was collected in 36.3 mL of 0.213 M HCl.

$$NH_3\ (g) + HCl\ (aq) \rightarrow NH_4Cl(aq)$$

This solution was titrated for excess HCl with 44.3 mL of 0.128 M NaOH. What is the percentage of nitrogen in the fertiliser ?

Solution :

Equivalent of acid for NH_3 ≡ equivalent of NH_3 from fertiliser

first we determine unreacted HCl (V mL)

V mL of 0.213 M HCl ≡ 44.3 mL of 0.128 M NaOH

$$V = \frac{44.3 \times 0.128}{0.213} = 26.62\,mL$$

used HCl = 46.30 – 26.62

= 19.68 mL

$$\text{amount of nitrogen in fertiliser} = \frac{NEV}{1000}$$

$$\text{equivalent to 19.68 mL of 0.213 M HCl} = \frac{0.213 \times 14 \times 19.68}{1000}\,g$$

= 0.0587 g

This amount is present in 0.608 g sample of fertilisers

$$\%\ \text{of nitrogen} = \frac{0.0587}{0.608} \times 100$$

= **9.65%**

Example 28

What mass of Na_2SO_3 must have been present in a sample that required 26.50 mL of 0.0510 M $KMnO_4$ for its oxidation to Na_2SO_4 in acidic solution ?

Solution :

Redox reaction is

$$5SO_3^{2-} + 2MnO_4^- + 6H^+ \rightarrow 2Mn^{2+} + 5S_4^{2-} + 3H_2O$$

$+ 5e^-$

$- 2e^-$

$$\text{equivalent weight of } MnO_4^- = \frac{M}{5}$$

$$\text{equivalent weight of } SO_3^{2-} = \frac{M}{2}$$

$$26.50 \text{ mL of } 0.510 \text{ M } MnO_4^- \equiv 26.50 \text{ mL of } 0.2550 \text{ N } MnO_4^-$$

$$\equiv 26.50 \text{ mL of } 0.255 \text{ N } SO_3^{2-}$$

$$\therefore \quad \text{weight of } Na_2SO_3 = \frac{NEV}{1000}$$

$$= \frac{0.2550 \times 63 \times 26.50}{100}$$

$$= \mathbf{0.426\ g}$$